

Compact-State Duality and Rigidity in Relativistic Free-Particle Dynamics

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Abstract We formulate classical relativistic free-particle dynamics in terms of a bounded vector–scalar state $(\mathbf{D}, S)$ satisfying $|\mathbf{D}|^2 + S^2 = 1$ with $S > 0$. A massive process carries a spatial displacement and an additive intrinsic scalar accumulation; the compact state specifies their normalized direction, while a positive radial coefficient defines the duration assigned by an inertial observer. This construction induces Lorentzian chronometry and identifies $S = d\tau/dt$. Requiring the same state to admit a canonical representation links its scalar component through $q = Q(d\tau/dt) = (H - \mathbf{p} \cdot \nabla H)/c$; when a regular Lagrangian description exists, the same scalar also satisfies $q = -L/c$. This yields a nonlinear consistency equation for an otherwise unspecified free Hamiltonian. We prove that a finite convex C^1 Hamiltonian with the calibrated rest value and a complete canonical representation of the massive state manifold is uniquely $H(\mathbf{p}) = c\sqrt{|\mathbf{p}|^2 + Q^2}$. Complementary radial and differential rigidity results give the same nondegenerate branch, whereas explicit projection solutions show why additional completeness or nondegeneracy assumptions are required. The resulting positive massive shell has $SO^+(1, 3)$ as its maximal connected linear covariance group; translation-origin freedom gives the proper orthochronous Poincaré group, and the compact boundary provides a projective null completion.

Keywords: motion–staticity state, relativistic free particles, Hamiltonian rigidity, convex duality, proper time, Lorentz covariance

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1 Introduction

The kinematics and dynamics of a relativistic free particle can be organized in several equivalent ways. Standard formulations begin with Lorentzian spacetime geometry, inertial-frame transformations, or the relativistic action and dispersion relation. Other reconstruction programs instead derive Lorentz kinematics from assumptions such as homogeneity, isotropy, reciprocity, continuity, or the composition law for inertial transformations [15, 1, 2, 5, 6, 7, 8]. The present problem is narrower. We ask how strongly free-particle dynamics is constrained if one bounded vector–scalar state is required to have mutually consistent operational and canonical representations.

The primitive massive state is

$$(\mathbf{D}, S) \in \mathbb{R}^3 \oplus \mathbb{R}, \quad |\mathbf{D}|^2 + S^2 = 1, \quad S > 0. \quad (1)$$

We refer to the vector and scalar sectors as motion and staticity, respectively. The quadratic relation in Eq. (1) is an internal Euclidean state-space normalization; it is not introduced as a spacetime metric. The spatial sector carries the ordinary $SO(3)$ action, whereas S is a spatial scalar.

Closely related velocity–proper-time parametrizations are already known. Brehme used the Euclidean relation between spatial velocity and proper-time rate in discussing relativistic velocity geometry [3], and Savasta and Di Stefano considered a unit vector involving the spatial velocity and $d\tau/dt$, together with a related Hamiltonian decomposition [11]. Hyperbolic and projective descriptions of relativistic velocity space are likewise standard [9, 4]. The unit relation itself, its hyperbolic image, and the usual proper-time formulas are therefore not claimed as new here. Related reconstructions of relativistic energy and momentum from kinematical consistency conditions were also developed by Sonogo and Pin [13, 14].

The distinction in the present construction is the role assigned to the bounded state. We do not begin by identifying $\mathbf{D}$ with $\mathbf{v}/c$ using a preexisting coordinate time. A massive process is instead assumed to carry a spatial displacement $d\mathbf{x}$ and an additive scalar accumulation $d\chi$. In one inertial operational representation the normalized state fixes the direction of this vector–scalar increment,

$$\left(\frac{d\mathbf{x}}{c}, d\chi \right) = dt (\mathbf{D}, S), \quad dt > 0. \quad (2)$$

Here dt is defined as the positive radial coefficient assigned to the process segment by that observer. Thus the compact state does not generate temporal extension by itself; an inertial state may remain

constant along an entire history. Rather, the state determines how spatial displacement and an additive scalar process variable combine into an observer-dependent radial duration.

Equation (2) gives

$$\mathbf{D} = \frac{\mathbf{v}}{c}, \quad \mathbf{v} := \frac{d\mathbf{x}}{dt}, \quad S = \frac{d\chi}{dt}. \quad (3)$$

Together with Eq. (1), it induces

$$c^2 d\chi^2 = c^2 dt^2 - d\mathbf{x}^2. \quad (4)$$

After this quadratic relation has been obtained we identify $d\tau := d\chi$. Hence

$$S = \frac{d\tau}{dt},$$

and the scalar accumulation acquires its standard proper-time interpretation. The construction therefore does not derive temporal ordering or intrinsic accumulation from the compact-state equation alone; it derives the relation between an additive massive-process accumulation, observer duration, and spatial displacement.

The same compact direction also carries a dimensional generator scale. For $Q > 0$, define

$$(\mathbf{d}, q) = Q(\mathbf{D}, S). \quad (5)$$

The two positive radial quantities have different roles: dt measures operational process extent in a chosen inertial representation, whereas Q is an intrinsic scale of the massive generator. In particular,

$$q = QS = Q \frac{d\tau}{dt}.$$

The central question arises when the same state is required to have a canonical representation. Let $H(\mathbf{p})$ be an initially unspecified differentiable autonomous free Hamiltonian. Hamilton's equation gives $\mathbf{v} = \nabla_{\mathbf{p}} H$. With

$$P_H = \left(\frac{H}{c}, \mathbf{p} \right), \quad \Pi_H = P_H^b = \left(-\frac{H}{c}, \mathbf{p} \right), \quad (6)$$

the canonical generator–translation pairing is

$$\Pi_H(dX) = \mathbf{p} \cdot d\mathbf{x} - H dt, \quad dX = (c dt, d\mathbf{x}).$$

Along a Hamiltonian trajectory this becomes

$$\Pi_H(dX) = -(H - \mathbf{p} \cdot \nabla H) dt.$$

Requiring the operational and Hamiltonian descriptions to represent the same scalar state component gives

$$q = Q \frac{d\tau}{dt} = \frac{H - \mathbf{p} \cdot \nabla H}{c}, \quad (7)$$

$$q dt = Q d\tau = -\frac{1}{c} \Pi_H(dX). \quad (8)$$

If a regular Lagrangian description exists, ordinary Legendre duality,

$$L = \mathbf{p} \cdot \mathbf{v} - H,$$

gives in addition

$$q = -\frac{L}{c} = \frac{H - \mathbf{p} \cdot \mathbf{v}}{c}.$$

The individual Legendre and proper-time identities are standard. The representation requirement is the equality of the operational and canonical scalar responses; the Lagrangian identity is an equivalent form when Legendre duality is available.

Together with the vector channel, this gives the common state representation

$$(\mathbf{D}, S) = \left(\frac{\mathbf{v}}{c}, \frac{d\tau}{dt} \right) = \left(\frac{\nabla H}{c}, \frac{H - \mathbf{p} \cdot \nabla H}{cQ} \right). \quad (9)$$

Imposing the compact normalization Eq. (1) then yields the central Hamiltonian consistency equation

$$\frac{|\nabla H|^2}{c^2} + \frac{(H - \mathbf{p} \cdot \nabla H)^2}{c^2 Q^2} = 1, \quad H - \mathbf{p} \cdot \nabla H > 0. \quad (10)$$

The inverse problem studied below is to determine when this state consistency fixes the free Hamiltonian.

Our strongest result uses canonical representation completeness. For a finite convex C^1 Hamiltonian with the calibrated rest value $H(\mathbf{0}) = cQ$, suppose Eq. (10) holds globally and the Hamiltonian representation realizes every massive compact state. Equivalently,

$$\nabla H(\mathbb{R}^3) = \{\mathbf{v} : |\mathbf{v}| < c\}. \quad (11)$$

We prove that these assumptions uniquely give

$$H(\mathbf{p}) = c \sqrt{|\mathbf{p}|^2 + Q^2}. \quad (12)$$

No independent isotropy assumption on H , strict convexity, or Hessian nondegeneracy is needed. We also prove two complementary rigidity results. In the radial C^2 class, the only global positive-branch solutions with $H(\mathbf{0}) = cQ$ are the constant branch and Eq. (12). Without radially, dense nondegeneracy of the Hessian again forces the same square-root Hamiltonian.

The additional hypotheses are substantive. If $\mathbf{R}$ is a proper orthogonal projection, then

$$H_{\mathbf{R}}(\mathbf{p}) = c \sqrt{Q^2 + |\mathbf{R}\mathbf{p}|^2} \quad (13)$$

satisfies the same pointwise consistency equation, but its state map covers only the sector $\mathbf{D} \in \text{Ran } \mathbf{R}$, and

$$\ker \mathbf{R} = \ker \nabla^2 H_{\mathbf{R}}.$$

These examples show that Eq. (10) alone can leave canonical directions invisible to the compact-state representation. Representation completeness and Legendre nondegeneracy eliminate different forms of this degeneracy.

The forward problem is naturally expressed by standard convex analysis. For a compact convex state body K with support function h_K , Legendre–Fenchel duality gives

$$H_{K,Q}(\mathbf{p}) = c h_K(\mathbf{p}, Q) \quad (14)$$

[10, 12]. Thus compactness alone does not select the relativistic square root: different state geometries generally generate different massive dispersion relations. Appendix B gives an L^r -type family for

which the conventional quadratic Newtonian limit selects the Euclidean member $r = 2$. For a general smooth state body, however, the Newtonian limit fixes only the local second-order curvature at the static state and not the full global geometry.

For the rigid Euclidean solution, writing

$$E := H(\mathbf{p}), \quad P = \left(\frac{E}{c}, \mathbf{p} \right),$$

gives

$$\eta(P, P) = -Q^2, \quad \eta = \text{diag}(-1, +1, +1, +1). \quad (15)$$

With $Q = mc$, this is the standard relation $E^2 - c^2|\mathbf{p}|^2 = m^2c^4$. The compact completion $S \geq 0$ is projectively equivalent to the future causal rays: $S > 0$ gives the timelike sector, while $S = 0$ gives the null boundary. A finite nonzero null momentum is obtained by the scale-free limit $Q \rightarrow 0^+$ at fixed nonzero momentum, rather than by the infinite-momentum limit on a fixed massive shell.

No Lorentz transformation law is assumed in obtaining the Hamiltonian rigidity result. Observer covariance is introduced only afterward. Operational translations form a vector space $\mathfrak{t}$, with canonical generator $\Pi \in \mathfrak{t}^*$. A continuous change of inertial description that preserves process addition acts linearly on $\mathfrak{t}$; invariance of the scalar pairing $\Pi(dX)$ induces the dual action on generators. Preservation of the positive massive shell then restricts this action to $O^\uparrow(1, 3)$, with connected component $SO^+(1, 3)$. Under comoving-frame completeness and unrestricted spatial reorientation, the complete proper orthochronous Lorentz group is realized. Allowing changes of event origin gives $\mathbb{R}^4 \rtimes SO^+(1, 3)$, the proper orthochronous Poincaré group.

This symmetry statement is distinct from the local construction of observer duration. The latter is defined by the radial realization Eq. (2); Lorentz covariance determines how the duration and spatial displacement assigned by one inertial observer are related to those assigned by another.

The scope of the result is correspondingly limited. The Euclidean compact state geometry, three-dimensional Euclidean spatial readout, additive massive-process scalar accumulation, and first-moment assumptions used in the operational cone construction are explicit inputs. The paper does not derive spacetime dimension, temporal ordering, or the entire structure of relativity from a structureless set of events. Its principal claim is instead an inverse dynamical one: once the bounded state is required to carry compatible operational and canonical representations, appropriate global completeness or differential nondegeneracy rigidly selects the standard relativistic free Hamiltonian.

The paper is organized as follows. Sec. 2 constructs the operational radial realization and the projective causal completion. Sec. 3 develops the canonical representation, convex duality, and the Hamiltonian rigidity results. Sec. 4 derives Lorentz and Poincaré covariance from the translation-generator structure. Sec. 5 discusses the assumptions, limitations, and possible extensions. The Appendixes supply the operational linear-extension argument, convex-dual examples, and the explicit degenerate projection family.

2 Compact State, Operational Duration, and Causal Completion

The primitive motion–staticity (MS) state is intended to describe the relative organization of motion and staticity without assuming a Lorentzian spacetime metric or an external coordinate time. We assume a three-dimensional Euclidean spatial readout and a class of inertial massive processes that can be described relationally by an observer. No absolute spatial rest frame or independently observable observer self-velocity is introduced.

A massive process is further assumed to carry an additive scalar accumulation, denoted initially by χ . Process concatenation is taken to have a chosen positive orientation, and for every nontrivial positively oriented segment,

$$d\chi > 0. \quad (16)$$

The increment $d\chi$ is taken to be attached to the massive process segment itself rather than to a particular inertial description; observer dependence enters through the spatial readout, the radial duration, and the corresponding motion–staticity decomposition. At this stage χ is only an extensive scalar process variable. No metric interpretation as proper time or observer duration is assigned to it.

2.1 Compact state and operational radial realization

Let

$$V = \mathbb{R}^3 \oplus \mathbb{R} \quad (17)$$

carry the spatial-rotation action

$$\rho(R)(\mathbf{d}, q) = (R\mathbf{d}, q), \quad R \in SO(3). \quad (18)$$

Thus $\mathbf{d}$ transforms as a spatial vector and q as a spatial scalar.

A positive-definite quadratic magnitude invariant under spatial rotations has, after independent positive calibration of its vector and scalar channels, the canonical form

$$|\mathbf{d}|^2 + q^2. \quad (19)$$

This is an internal state-space structure, not an assumed spacetime metric. For a massive system, let $Q > 0$ denote the fixed magnitude and impose

$$|\mathbf{d}|^2 + q^2 = Q^2, \quad q > 0. \quad (20)$$

Define

$$\mathbf{D} := \frac{\mathbf{d}}{Q}, \quad S := \frac{q}{Q}. \quad (21)$$

The normalized massive state manifold and its compact completion are

$$\mathcal{S} := \{(\mathbf{D}, S) : |\mathbf{D}|^2 + S^2 = 1, S > 0\}, \quad (22)$$

$$\overline{\mathcal{S}} := \{(\mathbf{D}, S) : |\mathbf{D}|^2 + S^2 = 1, S \geq 0\}. \quad (23)$$

The Euclidean geometry of $\mathcal{S}$ is a structural input of the present construction.

To separate normalized state direction from extensive process magnitude, let

$$C := \text{cone}(\mathcal{S}) \subset V \quad (24)$$

denote the positive cone generated by the massive states. The equivalent barycentric construction and its linear extension are given in Appendix A.

The operational first-moment readout is taken to have the form

$$\mathcal{R} : C \longrightarrow \mathbb{R}_{\text{disp}}^3 \oplus \mathbb{R}_{\chi}. \quad (25)$$

We impose barycentric sufficiency: preparations with the same unnormalized barycentric state have the same first-moment readout. We also assume additivity of this readout under combination of preparations and positive homogeneity under common positive rescaling. These assumptions make $\mathcal{R}$ a well-defined additive and positively homogeneous cone map, which extends uniquely to a real-linear map

$$\tilde{\mathcal{R}} : V \longrightarrow \mathbb{R}_{\text{disp}}^3 \oplus \mathbb{R}_\chi. \quad (26)$$

No linear structure is thereby imposed directly on the normalized pure-state manifold.

Spatial covariance requires

$$\tilde{\mathcal{R}}(R\mathbf{D}, S) = (R \oplus 1)\tilde{\mathcal{R}}(\mathbf{D}, S). \quad (27)$$

Since the vector and scalar sectors are inequivalent irreducible $SO(3)$ representations,

$$\tilde{\mathcal{R}}(\mathbf{D}, S) = (\alpha\mathbf{D}, \beta S) \quad (28)$$

for constants α, β . Positive process orientation fixes $\beta > 0$; nondegeneracy and orientation of the spatial channel allow us to take $\alpha > 0$.

We now work in one fixed inertial operational representation and suppress its observer label. Neither the normalized MS direction nor its positive radial coefficient is assumed to be observer invariant. A different inertial observer may assign both a different direction and a different radial coefficient to the same physical process segment; their relation will be determined only after the observer covariance structure has been constructed.

For a pure massive segment,

$$Y = \lambda(\mathbf{D}, S), \quad \lambda > 0, \quad (29)$$

the readout is

$$(d\mathbf{x}, d\chi) = \lambda(\alpha\mathbf{D}, \beta S). \quad (30)$$

The coefficient λ parametrizes the extensive size of the segment within the chosen operational representation.

Define the positive conversion scale and observer duration by

$$c := \frac{\alpha}{\beta}, \quad dt := \beta\lambda > 0. \quad (31)$$

At this stage c is the common calibration constant used to compare the spatial and scalar readout channels across the inertial operational representations considered below. No identification with the propagation speed of a particular field is assumed. Its role as the invariant causal scale, and its universality across massive species sharing the same observer structure, will follow from Sec. 4.

Equation (30) becomes

$$\left(\frac{d\mathbf{x}}{c}, d\chi\right) = dt(\mathbf{D}, S). \quad (32)$$

The normalized pair $(\mathbf{D}, S)$ therefore carries directional information, whereas dt is the extensive radial coefficient assigned by the observer.

Using $|\mathbf{D}|^2 + S^2 = 1$,

$$dt = \sqrt{d\chi^2 + \frac{|d\mathbf{x}|^2}{c^2}}. \quad (33)$$

For successive segments of the same inertial process, the common MS direction is fixed and the positive radial coefficients add. Thus t is an affine process parameter, unique up to an additive origin.

We may now define

$$\mathbf{v} := \frac{d\mathbf{x}}{dt}, \quad (34)$$

so that

$$\mathbf{D} = \frac{\mathbf{v}}{c}, \quad S = \frac{d\chi}{dt}. \quad (35)$$

For the pure-staticity state $(\mathbf{D}, S) = (\mathbf{0}, 1)$, Eq. (32) gives

$$d\mathbf{x} = 0, \quad dt = d\chi. \quad (36)$$

Thus the static state calibrates the scalar accumulation against the observer duration.

We assume operational completeness: every state in $\mathcal{S}$ is realized by an inertial massive process. Consequently the realized observer velocities fill the open ball

$$B_c := \{\mathbf{v} \in \mathbb{R}^3 : |\mathbf{v}| < c\}. \quad (37)$$

2.2 Lorentzian chronometry and projective causal completion

The compact-state relation and Eq. (35) imply

$$\frac{|\mathbf{v}|^2}{c^2} + \left(\frac{d\chi}{dt}\right)^2 = 1, \quad (38)$$

or equivalently

$$c^2 d\chi^2 = c^2 dt^2 - d\mathbf{x}^2. \quad (39)$$

Relations of this form are standard in relativistic velocity–proper-time parametrizations [3, 11]. In particular, the Euclidean unit relation between spatial velocity and proper-time rate is not claimed as new here. The role specific to the present construction is that observer duration has first been introduced as the radial coefficient of the operational MS realization rather than as an external coordinate-time parameter.

Introduce

$$dX := (c dt, d\mathbf{x}) \in \mathbb{R}^4 \quad (40)$$

and define

$$\eta := \text{diag}(-1, +1, +1, +1). \quad (41)$$

Equation (39) then reads

$$\eta(dX, dX) = -c^2 d\chi^2. \quad (42)$$

By operational completeness, the realized values of $\mathbf{D}$ fill the open unit ball, so the corresponding increments dX fill an open future-timelike cone in the translation space. Hence the operational relation determines the quadratic form uniquely: if another quadratic form agreed with Eq. (42) on all realized increments, the difference of the two quadratic forms would vanish on an open set and therefore vanish identically. Thus the operational radial relation uniquely induces the Lorentzian quadratic form on the four-dimensional translation coordinates. At this stage, however, invariance under changes of inertial observer has not been assumed or proved.

We now identify the intrinsic scalar increment with proper time by defining

$$d\tau := d\chi. \quad (43)$$

Hence τ and χ differ at most by an additive choice of origin, and

$$S = \frac{d\tau}{dt} = \sqrt{1 - \frac{|\mathbf{v}|^2}{c^2}}. \quad (44)$$

The Euclidean norm $|\mathbf{D}|^2 + S^2 = 1$ and the Lorentzian norm $\eta(dX, dX) = -c^2 d\tau^2$ therefore belong to different spaces: the former to the compact internal state geometry, the latter to operational translation space.

The compact state has the standard hyperbolic representation [3, 9, 4]. Let

$$\mathcal{H}^+ := \{U \in \mathbb{R}^4 : \eta(U, U) = -1, U^0 > 0\}. \quad (45)$$

For $S > 0$, define

$$\Phi : \mathcal{S} \longrightarrow \mathcal{H}^+, \quad \Phi(\mathbf{D}, S) = \left(\frac{1}{S}, \frac{\mathbf{D}}{S} \right). \quad (46)$$

Since

$$\eta(\Phi, \Phi) = \frac{-1 + |\mathbf{D}|^2}{S^2} = -1,$$

Φ maps $\mathcal{S}$ into the future unit hyperboloid. Its inverse is

$$\Phi^{-1}(U^0, \mathbf{U}) = \left(\frac{\mathbf{U}}{U^0}, \frac{1}{U^0} \right), \quad (47)$$

so the map is a diffeomorphism.

Its operational meaning follows directly from Eq. (32):

$$\Phi(\mathbf{D}, S) = \left(\frac{dt}{d\tau}, \frac{1}{c} \frac{d\mathbf{x}}{d\tau} \right). \quad (48)$$

After the Lorentzian interpretation has been made, this is the standard dimensionless future four-velocity. The four-velocity formula itself is standard; here Φ also records how the compact state is lifted by normalization with respect to intrinsic massive accumulation.

The compact boundary is most naturally described projectively. Let

$$C^+ := \{Y \in \mathbb{R}^4 \setminus \{0\} : Y^0 > 0, \eta(Y, Y) \leq 0\}, \quad (49)$$

and let $\mathbb{P}_+(C^+)$ denote the quotient by positive rescaling. Then

$$\widehat{\Phi} : \overline{\mathcal{S}} \longrightarrow \mathbb{P}_+(C^+), \quad \widehat{\Phi}(\mathbf{D}, S) = [(1, \mathbf{D})] \quad (50)$$

is a bijection. Indeed,

$$\eta((1, \mathbf{D}), (1, \mathbf{D})) = -1 + |\mathbf{D}|^2 = -S^2. \quad (51)$$

Thus $S > 0$ corresponds to future-timelike rays, whereas

$$S = 0, \quad |\mathbf{D}| = 1 \quad (52)$$

corresponds to future-null rays. Conversely, every future causal ray has a unique representative $(1, \mathbf{D})$ with $|\mathbf{D}| \leq 1$, and the corresponding compact-state coordinate is $S = \sqrt{1 - |\mathbf{D}|^2}$. Hence

$$\overline{\mathcal{S}} \simeq \mathbb{P}_+(C^+) \quad (53)$$

as compact projective state spaces.

For $S > 0$, Φ selects the unique future representative of Lorentzian norm -1 . At the null boundary this normalization ceases to exist, and

$$\frac{dt}{d\tau} = \frac{1}{S} \longrightarrow +\infty \quad (S \rightarrow 0^+). \quad (54)$$

The divergence is therefore a normalization effect, not a singularity of the projective causal ray.

Finally, the boundary $S = 0$ should not be produced by simply setting $Q = 0$ in $(\mathbf{D}, S) = (\mathbf{d}, q)/Q$, which is defined only for $Q > 0$. The finite-momentum null completion of the massive Hamiltonian family is described in Sec. 3.

The temporal hierarchy used below is therefore limited and explicit. The additive variable χ is a primitive scalar process accumulation; the MS direction determines how that scalar channel combines with spatial displacement; the positive radial coefficient defines an observer-dependent duration; and only then is $d\chi$ identified with the proper-time increment $d\tau$ of the induced Lorentzian geometry.

3 Compact-State Duality and Hamiltonian Rigidity

For a massive state,

$$(\mathbf{d}, q) = Q(\mathbf{D}, S), \quad Q > 0, \quad (55)$$

whereas the operational realization of Sec. 2 gives

$$S = \frac{d\tau}{dt}, \quad \mathbf{D} = \frac{\mathbf{v}}{c}.$$

Thus

$$q = QS = Q \frac{d\tau}{dt}. \quad (56)$$

The purpose of this section is to determine when the same compact state can be represented by an initially unspecified autonomous free Hamiltonian. No relativistic dispersion relation or four-momentum transformation law is assumed.

3.1 Canonical response and compact convex duality

Let

$$H : \mathbb{R}^3 \longrightarrow \mathbb{R} \quad (57)$$

be differentiable. Hamilton's equation gives

$$\mathbf{v} = \nabla_{\mathbf{p}} H. \quad (58)$$

Using the Lorentzian form induced in Sec. 2, introduce the generator vector candidate and its metric-dual action covector,

$$P_H := \left(\frac{H}{c}, \mathbf{p} \right), \quad \Pi_H := P_H^b = \left(-\frac{H}{c}, \mathbf{p} \right). \quad (59)$$

For

$$dX = (c dt, d\mathbf{x}),$$

the canonical pairing is

$$\Pi_H(dX) = \mathbf{p} \cdot d\mathbf{x} - H dt. \quad (60)$$

Along the Hamiltonian trajectory $d\mathbf{x} = \nabla H dt$,

$$\Pi_H(dX) = -(H - \mathbf{p} \cdot \nabla H) dt. \quad (61)$$

This defines the positive scalar canonical candidate

$$q_H := -\frac{\Pi_H(dX)}{c dt} = \frac{H - \mathbf{p} \cdot \nabla H}{c}. \quad (62)$$

If a regular Lagrangian description exists, ordinary Legendre duality gives

$$L = \mathbf{p} \cdot \mathbf{v} - H, \quad \mathbf{v} = \nabla H. \quad (63)$$

The MS representation-consistency requirement is

$$\mathbf{D}_H = \mathbf{D}, \quad q_H = q.$$

Consequently the scalar component has the equivalent representations

$$q = QS = Q \frac{d\tau}{dt} = -\frac{L}{c} = \frac{H - \mathbf{p} \cdot \mathbf{v}}{c} = \frac{H - \mathbf{p} \cdot \nabla H}{c}, \quad (64)$$

$$q dt = Q d\tau = -\frac{1}{c} \Pi_H(dX). \quad (65)$$

These two relations are the scalar representation bridge used throughout the remainder of the construction. Their individual Legendre and proper-time identities are standard; related velocity–proper-time decompositions were discussed, for example, by Savasta and Di Stefano [11]. The additional requirement here is that the operational, Lagrangian, and Hamiltonian quantities represent the same MS scalar component before the form of H has been fixed.

The vector and scalar components can therefore be written together as

$$(\mathbf{D}, S) = \left(\frac{\mathbf{v}}{c}, \frac{d\tau}{dt} \right) = \left(\frac{\nabla H}{c}, \frac{H - \mathbf{p} \cdot \nabla H}{cQ} \right). \quad (66)$$

The inverse problem of this paper is precisely the problem of determining which free Hamiltonians can realize Eq. (66) globally.

Imposing the compact-state normalization gives

$$\frac{|\nabla H|^2}{c^2} + \frac{(H - \mathbf{p} \cdot \nabla H)^2}{c^2 Q^2} = 1, \quad (67)$$

with the massive branch condition

$$H - \mathbf{p} \cdot \nabla H > 0. \quad (68)$$

Equivalently, Eq. (65) gives the generator–displacement relation

$$-\Pi_H(dX) = cQ d\tau. \quad (69)$$

The master equation is therefore a compact-state consistency condition, not an independent dispersion postulate.

For comparison, the forward construction is standard convex duality. Let $K \subset \mathbb{R}^3 \oplus [0, \infty)$ be compact and convex, with support function

$$h_K(\xi, \zeta) := \sup_{(\mathbf{D}, S) \in K} (\xi \cdot \mathbf{D} + \zeta S). \quad (70)$$

The corresponding Legendre–Fenchel construction gives

$$H_{K,Q}(\mathbf{p}) = c h_K(\mathbf{p}, Q), \quad (71)$$

a standard support-function identity [10, 12]. Details and non-Euclidean examples are collected in Appendix B.

For the Euclidean state body

$$K_2^+ := \text{conv } \bar{\mathcal{S}} = \{(\mathbf{D}, S) : |\mathbf{D}|^2 + S^2 \leq 1, S \geq 0\}, \quad (72)$$

one obtains

$$h_{K_2^+}(\mathbf{p}, Q) = \sqrt{|\mathbf{p}|^2 + Q^2}, \quad H_Q(\mathbf{p}) = c \sqrt{|\mathbf{p}|^2 + Q^2}. \quad (73)$$

The associated closed convex Lagrangian is

$$L_Q(\mathbf{v}) = \begin{cases} -cQ \sqrt{1 - \frac{|\mathbf{v}|^2}{c^2}}, & |\mathbf{v}| \leq c, \\ +\infty, & |\mathbf{v}| > c. \end{cases} \quad (74)$$

For $Q > 0$, the boundary value $|\mathbf{v}| = c$ belongs to the closed convex extension and is not a finite-momentum massive state.

This forward construction also shows that compactness alone does not select the relativistic square root: different compact bodies generally have different support functions. The rigidity question is the converse one.

3.2 Hamiltonian rigidity and causal completion

Definition 3.1 (Canonical representation completeness). *A Hamiltonian satisfying Eqs. (67) and (68) is representation-complete if*

$$\Psi_H(\mathbf{p}) := \left(\frac{\nabla H(\mathbf{p})}{c}, \frac{H(\mathbf{p}) - \mathbf{p} \cdot \nabla H(\mathbf{p})}{cQ} \right) \quad (75)$$

maps $\mathbb{R}^3$ onto $\mathcal{S}$.

Because the positive scalar component is fixed by the vector component, representation completeness is equivalent to

$$\nabla H(\mathbb{R}^3) = B_c = \{\mathbf{v} : |\mathbf{v}| < c\}. \quad (76)$$

Theorem 3.2 (Rigidity from complete MS representation). *Let $H : \mathbb{R}^3 \rightarrow \mathbb{R}$ be finite, convex, and C^1 . Suppose*

$$H(\mathbf{0}) = cQ, \quad c > 0, \quad Q > 0, \quad (77)$$

Eqs. (67) and (68) hold globally, and H is representation-complete. Then

$$H(\mathbf{p}) = c \sqrt{|\mathbf{p}|^2 + Q^2}. \quad (78)$$

Proof. At $\mathbf{p} = \mathbf{0}$, the master equation and $H(\mathbf{0}) = cQ$ imply $\nabla H(\mathbf{0}) = 0$. Convexity therefore makes $\mathbf{0}$ a global minimizer.

Let

$$H^*(\mathbf{v}) := \sup_{\mathbf{p}} (\mathbf{p} \cdot \mathbf{v} - H(\mathbf{p})) \quad (79)$$

be the convex conjugate. For $|\mathbf{v}| < c$, representation completeness gives a momentum with $\mathbf{v} = \nabla H(\mathbf{p})$. Fenchel equality and the positive branch of Eq. (67) then give

$$H^*(\mathbf{v}) = -cQ \sqrt{1 - \frac{|\mathbf{v}|^2}{c^2}}, \quad |\mathbf{v}| < c. \quad (80)$$

The positive branch also implies $|\nabla H| < c$. Integration along line segments yields

$$|H(\mathbf{p}) - H(\mathbf{q})| \leq c|\mathbf{p} - \mathbf{q}|, \quad (81)$$

so H is globally c -Lipschitz. If $|\mathbf{v}| > c$, taking $\mathbf{p} = t\mathbf{v}/|\mathbf{v}|$ gives

$$\mathbf{p} \cdot \mathbf{v} - H(\mathbf{p}) \geq t(|\mathbf{v}| - c) - H(\mathbf{0}) \longrightarrow +\infty,$$

and hence

$$H^*(\mathbf{v}) = +\infty, \quad |\mathbf{v}| > c. \quad (82)$$

It remains to determine the boundary value. Let $|\mathbf{v}_0| = c$ and $\mathbf{v}_s = s\mathbf{v}_0$, $0 < s < 1$. From Eq. (80),

$$H^*(\mathbf{v}_s) = -cQ \sqrt{1 - s^2} \longrightarrow 0.$$

Lower semicontinuity gives $H^*(\mathbf{v}_0) \leq 0$. Since $H^*(\mathbf{0}) = -cQ$, convexity gives

$$H^*(\mathbf{v}_s) \leq sH^*(\mathbf{v}_0) + (1 - s)H^*(\mathbf{0}),$$

and therefore $H^*(\mathbf{v}_0) \geq 0$ as $s \rightarrow 1^-$. Thus

$$H^*(\mathbf{v}) = \begin{cases} -cQ \sqrt{1 - \frac{|\mathbf{v}|^2}{c^2}}, & |\mathbf{v}| \leq c, \\ +\infty, & |\mathbf{v}| > c. \end{cases} \quad (83)$$

A finite convex function on $\mathbb{R}^3$ is closed, hence $H = H^{**}$. Consequently

$$\begin{aligned} H(\mathbf{p}) &= \sup_{|\mathbf{v}| \leq c} \left[\mathbf{p} \cdot \mathbf{v} + cQ \sqrt{1 - \frac{|\mathbf{v}|^2}{c^2}} \right] \\ &= c \sqrt{|\mathbf{p}|^2 + Q^2}, \end{aligned} \quad (84)$$

the support function of the Euclidean unit half-ball. $\square$

The theorem requires no independent isotropy assumption on H , no strict convexity, and no Hessian nondegeneracy. Its additional input is global: the canonical representation must cover the complete massive MS state manifold.

Two complementary results replace this global hypothesis by different regularity assumptions.

Theorem 3.3 (Radial rigidity). *Let $H \in C^2(\mathbb{R}^3)$ be radial,*

$$H(\mathbf{p}) = h(r), \quad r = |\mathbf{p}|,$$

with $H(\mathbf{0}) = cQ$. If Eqs. (67) and (68) hold globally, then either

$$H \equiv cQ \tag{85}$$

or

$$H(\mathbf{p}) = c \sqrt{|\mathbf{p}|^2 + Q^2}. \tag{86}$$

Proof. Set $e = h/c$. Radial C^2 regularity gives $e'(0) = 0$, while the master equation becomes

$$e'^2 + \frac{(e - re')^2}{Q^2} = 1, \quad e - re' > 0. \tag{87}$$

There is therefore a C^1 function $\theta \in (-\pi/2, \pi/2)$ such that

$$e' = \sin \theta, \quad \frac{e - re'}{Q} = \cos \theta, \quad \theta(0) = 0.$$

Differentiation gives

$$(r \cos \theta - Q \sin \theta) \theta' = 0. \tag{88}$$

Let $U = \{r > 0 : \theta'(r) \neq 0\}$. On every connected component of U ,

$$\theta(r) = \arctan(r/Q), \quad \theta'(r) = \frac{Q}{r^2 + Q^2} > 0.$$

Continuity of θ' excludes any finite endpoint of such a component. Hence either $U = \emptyset$ or $U = (0, \infty)$. In the first case $\theta \equiv 0$, so $H \equiv cQ$. In the second,

$$e'(r) = \frac{r}{\sqrt{r^2 + Q^2}},$$

and $e(0) = Q$ gives $e(r) = \sqrt{r^2 + Q^2}$. □

Theorem 3.4 (Differential rigidity). *Let $H \in C^2(\mathbb{R}^3)$ satisfy Eqs. (67) and (68), with $H(\mathbf{0}) = cQ$. If*

$$\det \nabla^2 H \neq 0 \tag{89}$$

on a dense subset of $\mathbb{R}^3$, then

$$H(\mathbf{p}) = c \sqrt{|\mathbf{p}|^2 + Q^2}. \tag{90}$$

Proof. Set

$$G := H - \mathbf{p} \cdot \nabla H.$$

Then $\nabla G = -\nabla^2 H \mathbf{p}$. Differentiating Eq. (67) yields

$$\nabla^2 H \left(\nabla H - \frac{G}{Q^2} \mathbf{p} \right) = 0. \tag{91}$$

On the dense set where the Hessian is invertible,

$$\nabla H = \frac{G}{Q^2} \mathbf{p}.$$

Continuity extends this equality to all $\mathbf{p}$. Substitution in the master equation and the condition $G > 0$ give

$$G = \frac{cQ^2}{\sqrt{|\mathbf{p}|^2 + Q^2}}, \quad \nabla H = \frac{c\mathbf{p}}{\sqrt{|\mathbf{p}|^2 + Q^2}}. \quad (92)$$

Integration and $H(\mathbf{0}) = cQ$ complete the proof. $\square$

The extra hypotheses cannot simply be omitted. Let $\mathbf{R}$ be an orthogonal projection and define

$$H_{\mathbf{R}}(\mathbf{p}) = c \sqrt{Q^2 + |\mathbf{R}\mathbf{p}|^2}. \quad (93)$$

Then

$$\nabla H_{\mathbf{R}} = \frac{c \mathbf{R}\mathbf{p}}{\sqrt{Q^2 + |\mathbf{R}\mathbf{p}|^2}}, \quad H_{\mathbf{R}} - \mathbf{p} \cdot \nabla H_{\mathbf{R}} = \frac{cQ^2}{\sqrt{Q^2 + |\mathbf{R}\mathbf{p}|^2}}, \quad (94)$$

so Eq. (67) holds exactly. If $\mathbf{R} \neq I$, however,

$$\nabla H_{\mathbf{R}}(\mathbb{R}^3) \subset \text{Ran } \mathbf{R}, \quad \ker \mathbf{R} \subseteq \ker \nabla^2 H_{\mathbf{R}},$$

and both representation completeness and dense Hessian nondegeneracy fail. These MS-invisible canonical directions are analyzed in Appendix C; no gauge interpretation of them is assumed.

We now return to the nondegenerate branch,

$$H_Q(\mathbf{p}) = c \sqrt{|\mathbf{p}|^2 + Q^2}. \quad (95)$$

Its canonical MS state is

$$\mathbf{D}_H = \frac{\mathbf{p}}{\sqrt{|\mathbf{p}|^2 + Q^2}}, \quad S_H = \frac{Q}{\sqrt{|\mathbf{p}|^2 + Q^2}}. \quad (96)$$

Writing

$$E := H_Q(\mathbf{p}), \quad P := \left(\frac{E}{c}, \mathbf{p} \right), \quad \Pi := P^b = \left(-\frac{E}{c}, \mathbf{p} \right), \quad (97)$$

the projective map satisfies

$$P = Q \Phi(\mathbf{D}_H, S_H). \quad (98)$$

Therefore

$$\eta(P, P) = -Q^2, \quad P^0 > 0. \quad (99)$$

With

$$m := \frac{Q}{c}, \quad (100)$$

this is

$$E^2 - c^2 |\mathbf{p}|^2 = m^2 c^4. \quad (101)$$

The scalar bridge simultaneously reduces to the standard action relation

$$\Pi(dX) = -mc^2 d\tau. \quad (102)$$

The compact boundary also admits a finite-momentum null completion. At fixed $\mathbf{p} \neq 0$,

$$\lim_{Q \rightarrow 0^+} H_Q(\mathbf{p}) = c|\mathbf{p}|, \quad (\mathbf{D}_H, S_H) \longrightarrow \left(\frac{\mathbf{p}}{|\mathbf{p}|}, 0 \right). \quad (103)$$

Thus

$$H_0(\mathbf{p}) = c|\mathbf{p}|, \quad \mathbf{p} \neq 0, \quad (104)$$

with $|\mathbf{v}| = c$ and vanishing scalar MS component. The convex-dual interpretation of this limit is given in Appendix B.

A nonzero null momentum is naturally a homogeneous lift of the projective boundary:

$$P = \varrho(1, \mathbf{D}), \quad \varrho > 0, \quad |\mathbf{D}| = 1. \quad (105)$$

Then

$$\eta(P, P) = 0. \quad (106)$$

In the massive interior,

$$P = \frac{Q}{S}(1, \mathbf{D}), \quad \varrho = \frac{Q}{S}, \quad (107)$$

whereas at $S = 0$ the positive null scale ϱ is independent of the vanished massive normalization Q .

The two approaches to the projective boundary are therefore distinct:

$$\begin{aligned} Q > 0, \quad S \rightarrow 0^+ &\implies |\mathbf{p}| \rightarrow \infty && \text{on a fixed massive shell,} \\ Q \rightarrow 0^+, \quad \mathbf{p} \neq 0 \text{ fixed} &\implies S \rightarrow 0 && \text{with finite null momentum.} \end{aligned} \quad (108)$$

The structure selected in this section can finally be summarized by the two radial realizations

$$\left(\frac{d\mathbf{x}}{c}, d\tau \right) = dt(\mathbf{D}, S), \quad (\mathbf{d}, q) = Q(\mathbf{D}, S), \quad (109)$$

together with the scalar incidence relation

$$q dt = Q d\tau = -\frac{1}{c}\Pi(dX).$$

Canonical representation consistency requires both radial realizations to carry the same normalized MS state. Under complete representation, or under the complementary regularity assumptions above, this requirement selects the standard relativistic free Hamiltonian.

No observer transformation law has been used in obtaining this result. The next section begins with the dynamically selected massive shell

$$\eta(P, P) = -Q^2, \quad P^0 > 0,$$

and determines its maximal connected covariance group and the conditions under which that covariance is realized by inertial observers.

4 Lorentz and Poincaré Structure

Sections 2 and 3 produced two structures without assuming a Lorentz transformation law between observers. The operational radial realization induced the quadratic form

$$\eta = \text{diag}(-1, +1, +1, +1), \quad (110)$$

while canonical representation consistency and Hamiltonian rigidity selected the positive massive shell

$$\mathcal{H}_Q^+ := \{P \in \mathbb{R}^4 : \eta(P, P) = -Q^2, P^0 > 0\}, \quad Q > 0. \quad (111)$$

Here

$$P = \left(\frac{E}{c}, \mathbf{p}\right), \quad \Pi := P^\flat = \left(-\frac{E}{c}, \mathbf{p}\right). \quad (112)$$

We now ask how two inertial operational descriptions of the same process can be related. No absolute observer velocity is introduced. The Lorentz and Poincaré groups obtained below are standard structures of relativistic kinematics [1, 2, 4]; the issue here is their place in the preceding reconstruction.

4.1 Translation duality and Lorentz covariance

Let

$$\mathfrak{t} \simeq \mathbb{R}^4 \quad (113)$$

denote the real vector space generated by operational translation increments,

$$dX = (c \, dt, d\mathbf{x}) \in \mathfrak{t}. \quad (114)$$

The canonical generator belongs naturally to the dual space,

$$\Pi \in \mathfrak{t}^*, \quad \Pi(dX) = \mathbf{p} \cdot d\mathbf{x} - E \, dt. \quad (115)$$

Thus the generator–translation pairing is defined before any use is made of a Lorentz transformation law.

We first record the maximal linear covariance of the massive shell.

Theorem 4.1 (Linear automorphisms of the positive massive shell). *The real-linear automorphism group of $\mathcal{H}_Q^+$ is*

$$\text{Aut}_{\text{lin}}(\mathcal{H}_Q^+) = O^\uparrow(1, 3), \quad (116)$$

and its identity component is

$$\text{Aut}_{\text{lin}}^0(\mathcal{H}_Q^+) = SO^+(1, 3). \quad (117)$$

Proof. Let $A \in GL(4, \mathbb{R})$ map $\mathcal{H}_Q^+$ onto itself and set

$$C := A^T \eta A - \eta.$$

Using the parametrization

$$P(\mathbf{p}) = \left(\sqrt{Q^2 + |\mathbf{p}|^2}, \mathbf{p}\right),$$

shell preservation implies

$$P(\mathbf{p})^T C P(\mathbf{p}) = 0 \quad \text{for all } \mathbf{p} \in \mathbb{R}^3. \quad (118)$$

At $\mathbf{p} = \mathbf{0}$ this gives $C_{00} = 0$. Differentiating with respect to p_i at the origin gives $C_{0i} = 0$. Equation (118) then reduces to

$$C_{ij} p^i p^j = 0 \quad \text{for all } \mathbf{p},$$

so $C_{ij} = 0$ and therefore

$$A^T \eta A = \eta.$$

Because A preserves the positive sheet, $A \in O^\uparrow(1, 3)$. The converse is immediate, and the identity component is $SO^+(1, 3)$. $\square$

Now consider two inertial operational representations A and B . Their translation increments are related by a continuous invertible map

$$M_{BA} : \mathfrak{t} \longrightarrow \mathfrak{t}. \quad (119)$$

Preservation of process concatenation gives

$$M_{BA}(X + Y) = M_{BA}X + M_{BA}Y. \quad (120)$$

A continuous additive map between finite-dimensional real vector spaces is real linear. This linearity follows from process composition and does not assume Lorentz covariance.

The scalar generator–translation incidence must describe the same physical process in either representation:

$$\Pi_B(M_{BA}dX_A) = \Pi_A(dX_A). \quad (121)$$

Hence

$$\Pi_B = M_{BA}^{-T} \Pi_A. \quad (122)$$

Using $P = \eta^\#(\Pi)$, the associated generator-vector transformation is

$$P_B = A_{BA}P_A, \quad A_{BA} := \eta M_{BA}^{-T} \eta. \quad (123)$$

Suppose the two observers describe the same massive species and therefore the same intrinsic scale Q . Then A_{BA} maps $\mathcal{H}_Q^+$ onto itself. By Theorem 4.1,

$$A_{BA}^T \eta A_{BA} = \eta.$$

For a Lorentz matrix,

$$A_{BA}^{-T} = \eta A_{BA} \eta.$$

Equation (123) also gives $M_{BA}^{-T} = \eta A_{BA} \eta$, and therefore

$$M_{BA} = A_{BA}. \quad (124)$$

Thus the same Lorentz matrix acts on translation increments and generator vectors:

$$dX_B = \Lambda_{BA} dX_A, \quad P_B = \Lambda_{BA} P_A, \quad \Lambda_{BA} \in O^\uparrow(1, 3). \quad (125)$$

For observer changes continuously connected to the identity,

$$\Lambda_{BA} \in SO^+(1, 3). \quad (126)$$

This is the first stage at which transformations between observer-dependent durations enter. Each observer has already assigned its own radial duration through the operational construction of Sec. 2; Lorentz covariance determines the relation between those assignments.

4.2 Observer realization, universal scale, and affine completion

Let G_{obs} be a connected Lie group of inertial observer changes and let

$$\Lambda : G_{\text{obs}} \longrightarrow SO^+(1, 3) \quad (127)$$

be the smooth representation obtained above. The shell argument constrains the observer image but does not by itself imply that every Lorentz transformation is physically realized.

Corollary 4.2 (Full connected Lorentz realization). *Suppose that $\Lambda(G_{\text{obs}})$ contains the full spatial rotation group $SO(3)$ and acts transitively on $\mathcal{H}_Q^+$. Then*

$$\Lambda(G_{\text{obs}}) = SO^+(1, 3). \quad (128)$$

If Λ is faithful, then $G_{\text{obs}} \simeq SO^+(1, 3)$.

Proof. Set

$$H_{\text{obs}} := \Lambda(G_{\text{obs}}) \subseteq SO^+(1, 3).$$

Transitivity on the three-dimensional hyperboloid implies a three-dimensional orbit. The stabilizer of $P_0 = (Q, \mathbf{0})$ contains $SO(3)$, so it has dimension at least three. Hence

$$\dim H_{\text{obs}} \geq 6.$$

Since $\dim SO^+(1, 3) = 6$, the two Lie algebras coincide. Thus H_{obs} is open in the connected group $SO^+(1, 3)$, and consequently $H_{\text{obs}} = SO^+(1, 3)$. $\square$

The transitivity assumption is equivalent to comoving-frame completeness: every massive inertial state admits an observer for which its generator is represented by $P_0 = (Q, \mathbf{0})$. Full Lorentz realization therefore requires more than invariance of the shell itself.

We next address the conversion scale c . Let two massive species A and B be described using common physical units for spatial displacement and observer duration, while allowing a priori distinct conversion constants c_A and c_B . Suppose their generators transform under the same observer action and that this action contains a transformation mixing temporal and spatial directions.

Proposition 4.3 (Universality of the conversion scale). *If the massive shell of each species is invariant under the common observer action, then*

$$c_A = c_B. \quad (129)$$

Proof. Choose an arbitrary reference constant $c_0 > 0$ and write

$$Y = \left(\frac{E}{c_0}, \mathbf{p} \right).$$

For species $s \in \{A, B\}$, its shell is

$$Y^T G_s Y = -Q_s^2, \quad G_s = \text{diag} \left(-\frac{c_0^2}{c_s^2}, 1, 1, 1 \right). \quad (130)$$

By the shell-rigidity argument of Theorem 4.1, invariance of each nonzero massive hyperboloid implies

$$B_g^T G_s B_g = G_s$$

for every common observer transformation B_g . It follows that B_g commutes with

$$G_A^{-1}G_B = \text{diag}\left(\frac{c_A^2}{c_B^2}, 1, 1, 1\right). \quad (131)$$

If $c_A \neq c_B$, the one-dimensional temporal eigenspace and the three-dimensional spatial eigenspace are distinct and must be preserved separately by every B_g . No common transformation could then mix temporal and spatial directions. Hence $c_A = c_B$. $\square$

Thus a common nontrivial relativistic observer structure makes the local conversion scale species independent. The same c defines the null boundary preserved by the resulting Lorentz group. Identifying this causal scale with the propagation speed of a particular massless field requires the additional physical statement that the field realizes this null sector.

Once the event set is organized as an affine space modeled on the translation vector space $\mathfrak{t}$, freedom in the choice of origin gives

$$X' = \Lambda X + a, \quad a \in \mathfrak{t}. \quad (132)$$

The maximal connected affine covariance group is therefore

$$\mathcal{P}_+^\uparrow = \mathfrak{t} \rtimes SO^+(1, 3) \simeq \mathbb{R}^4 \rtimes SO^+(1, 3), \quad (133)$$

the proper orthochronous Poincaré group.

Finally, the linear Lorentz action induces the familiar nonlinear action on the bounded MS variables. Put

$$Y = (1, \mathbf{D})$$

and, for $\Lambda \in SO^+(1, 3)$, define

$$a(\Lambda, \mathbf{D}) := \Lambda^0_0 + \Lambda^0_j D^j. \quad (134)$$

For every future causal state $a(\Lambda, \mathbf{D}) > 0$. Writing

$$\Lambda(1, \mathbf{D}) = a(\Lambda, \mathbf{D})(1, \mathbf{D}')$$

and using preservation of $\eta((1, \mathbf{D}), (1, \mathbf{D})) = -S^2$ gives

$$D'^i = \frac{\Lambda^i_0 + \Lambda^i_j D^j}{\Lambda^0_0 + \Lambda^0_j D^j}, \quad (135)$$

$$S' = \frac{S}{\Lambda^0_0 + \Lambda^0_j D^j}. \quad (136)$$

For $S > 0$, equivalently,

$$(\mathbf{D}', S') = \Phi^{-1} \circ \Lambda \circ \Phi(\mathbf{D}, S). \quad (137)$$

With $\mathbf{D} = \mathbf{v}/c$, Eq. (135) is the standard relativistic velocity transformation law.

The same action extends continuously to the null boundary. For

$$P = \varrho(1, \mathbf{D}), \quad \varrho > 0, \quad |\mathbf{D}| = 1, \quad (138)$$

one has

$$P' = \varrho'(1, \mathbf{D}'), \quad \varrho' = a(\Lambda, \mathbf{D})\varrho. \quad (139)$$

Thus Lorentz transformations act projectively on the compact causal direction and homogeneously on the positive null generator scale.

The result of this section is therefore a separation between maximal covariance and physical realization. The massive shell fixes $SO^+(1, 3)$ as the maximal connected linear covariance group, whereas the realization of the full group by inertial observers requires comoving-frame completeness and unrestricted spatial reorientation. Observer duration is defined locally before this step; Lorentz covariance determines how the durations and spatial displacements assigned by different inertial observers are related.

5 Discussion and Outlook

We have formulated relativistic free-particle dynamics in terms of a bounded motion–staticity state

$$(\mathbf{D}, S), \quad |\mathbf{D}|^2 + S^2 = 1, \quad S > 0,$$

and required the same state to admit compatible operational and canonical representations. The familiar proper-time relation, future unit hyperboloid, relativistic mass shell, Lorentz group, Poincaré group, and Legendre–Fenchel duality are standard structures [3, 9, 4, 10, 12]. The main content of the present work is the representation-consistency problem that links them and the rigidity results that follow from it.

The temporal assumptions should be stated precisely. The instantaneous compact state does not generate a time parameter by itself: a free inertial state can remain constant along an entire history. The construction therefore assumes a positive additive scalar process accumulation $d\chi$ and a spatial process increment. Their observer-dependent radial realization defines dt , after which the induced Lorentzian relation permits the identification $d\tau = d\chi$. Thus coordinate duration is not introduced as an external Newtonian parameter, but neither temporal ordering nor the existence of intrinsic process accumulation is derived from $|\mathbf{D}|^2 + S^2 = 1$ alone. An entirely internal realization of the observer label, in which the operational reference is itself a physical process belonging to the same closed system, is given in Appendix D. This internal realization does not alter the temporal assumptions stated above.

The central dynamical bridge is the identification of the same MS scalar component in the operational and canonical representations,

$$q = Q \frac{d\tau}{dt} = \frac{H - \mathbf{p} \cdot \nabla H}{c}, \quad q dt = Q d\tau = -\frac{1}{c} \Pi(dX). \quad (140)$$

When a regular Lagrangian description exists, Legendre duality gives the additional equivalent representation

$$q = -\frac{L}{c} = \frac{H - \mathbf{p} \cdot \mathbf{v}}{c}.$$

The individual proper-time and Legendre identities are standard. The additional requirement is that the operational and canonical quantities describe the same scalar component of one compact state while $\mathbf{D} = \mathbf{v}/c = \nabla H/c$. Combining this common representation with the compact normalization produces the Hamiltonian consistency equation

$$\frac{|\nabla H|^2}{c^2} + \frac{(H - \mathbf{p} \cdot \nabla H)^2}{c^2 Q^2} = 1, \quad H - \mathbf{p} \cdot \nabla H > 0. \quad (141)$$

The strongest inverse result shows that a finite convex C^1 Hamiltonian with $H(\mathbf{0}) = cQ$, satisfying Eq. (141) globally and representing the complete massive MS manifold, is necessarily

$$H(\mathbf{p}) = c \sqrt{|\mathbf{p}|^2 + Q^2}. \quad (142)$$

No independent Hamiltonian isotropy, strict convexity, or Hessian nondegeneracy is required. The radial and differential rigidity theorems provide complementary sufficient conditions. Conversely, the projection family of Sec. 3 and Appendix C shows that the pointwise consistency equation alone permits lower-dimensional canonical realizations. The additional hypotheses therefore remove genuine representation degeneracies rather than merely technical irregularities.

The forward convex construction gives the complementary limitation. For a compact convex state body K ,

$$H_{K,Q}(\mathbf{p}) = c h_K(\mathbf{p}, Q),$$

so different state geometries generally give different massive dispersion relations. The Euclidean geometry of the primitive MS state is therefore a structural input of this paper, not a consequence of Hamiltonian consistency. The Newtonian quadratic limit constrains its local curvature at the static state but, for a general compact body, does not determine its global shape.

For the Euclidean state body the rigid Hamiltonian yields

$$P = \left(\frac{E}{c}, \mathbf{p} \right), \quad \eta(P, P) = -Q^2,$$

and, with $Q = mc$, the standard massive dispersion relation follows. The projective boundary $S = 0$ represents null directions. Two limits must nevertheless be distinguished: $S \rightarrow 0^+$ at fixed $Q > 0$ is an infinite-momentum limit on one massive shell, whereas $Q \rightarrow 0^+$ at fixed nonzero momentum produces the finite null dispersion $E = c|\mathbf{p}|$. This construction identifies a free null sector, not the dynamics of any particular massless field.

The symmetry result has a similarly restricted meaning. The positive mass shell has $SO^+(1, 3)$ as its maximal connected linear covariance group, and translation-origin freedom gives the proper orthochronous Poincaré group. A physical family of inertial observers need not realize the entire Lorentz group unless additional completeness assumptions are made. In the present framework the Lorentz transformation does not create observer duration; it relates the different radial durations and spatial increments assigned to the same process by different inertial observers.

Several assumptions therefore remain explicit. We assume a three-dimensional Euclidean spatial readout and its rotational organization, the Euclidean compact-state norm, additive intrinsic accumulation along massive processes, and the first-moment conditions used to obtain the linear operational readout. The construction is not a derivation of spacetime dimension or of all relativistic structure from a structureless set of events. Its narrower claim is that these inputs, together with common operational-canonical representation and suitable completeness or nondegeneracy conditions, rigidly select the standard massive free Hamiltonian and its Lorentz–Poincaré covariance.

The generator–translation pairing

$$\Pi(dX) = \mathbf{p} \cdot d\mathbf{x} - E dt$$

is particularly useful for extensions because it does not require one to regard momentum primarily as a coordinate four-vector. In the present flat problem the translation space is global and the metric identifies generator vectors and covectors. In a curved geometry there is generally no canonical global

translation group or path-independent addition of local momenta. A natural extension is therefore to investigate local generator–displacement response before introducing any global momentum notion.

Interactions raise a separate issue. The free-particle relation

$$-\Pi(dX) = cQ d\tau$$

does not determine the generator of an interacting or composite system, whose energy and momentum include interaction contributions. Spin, internal charges, fields, quantum systems, and gravitation likewise require structures absent from the present analysis.

The result should consequently be read as a reconstruction and rigidity statement for classical relativistic free-particle dynamics. Its central structure is the compatibility

compact MS state $\longleftrightarrow$ operational process $\longleftrightarrow$ canonical generator $\longleftrightarrow$ relational observer description.

Under the hypotheses stated above, this compatibility selects the standard relativistic massive Hamiltonian and the connected Lorentz–Poincaré covariance structure.

A Operational Cone and Linear Extension

This appendix gives the linear-extension argument used in Sec. 2. The normalized massive state manifold is

$$\mathcal{S} = \{(\mathbf{D}, S) \in \mathbb{R}^3 \oplus \mathbb{R} : |\mathbf{D}|^2 + S^2 = 1, S > 0\}. \quad (143)$$

No linear structure is imposed directly on $\mathcal{S}$. Positive scaling is introduced only after passing to unnormalized process or ensemble data.

Let

$$V := \mathbb{R}^3 \oplus \mathbb{R}.$$

For a finite positive measure μ on $\mathcal{S}$, define its unnormalized barycentric state by

$$Y_\mu := \int_{\mathcal{S}} X d\mu(X), \quad X = (\mathbf{D}, S), \quad (144)$$

and let

$$C := \{Y_\mu : \mu \text{ is a finite positive measure on } \mathcal{S}\}. \quad (145)$$

Equivalently, C is the positive cone generated by $\mathcal{S}$.

Suppose a preparation μ has first-moment operational readout

$$\mathcal{M}(\mu) \in W, \quad W := \mathbb{R}_{\text{disp}}^3 \oplus \mathbb{R}_\chi. \quad (146)$$

We assume barycentric sufficiency,

$$Y_\mu = Y_\nu \implies \mathcal{M}(\mu) = \mathcal{M}(\nu), \quad (147)$$

together with additivity under combination of preparations and positive homogeneity under common positive rescaling. Hence

$$\mathcal{R}(Y_\mu) := \mathcal{M}(\mu) \quad (148)$$

defines an additive, positively homogeneous map $\mathcal{R} : C \rightarrow W$.

Lemma A.1. *The cone C generates the whole ambient space:*

$$C - C = V. \quad (149)$$

Proof. The static state

$$X_0 = (\mathbf{0}, 1)$$

belongs to $\mathcal{S}$. For any $a \in (0, 1)$ and standard basis vectors $\mathbf{e}_i$, the states

$$X_i^\pm = (\pm a\mathbf{e}_i, \sqrt{1-a^2})$$

also belong to $\mathcal{S}$, and

$$X_i^+ - X_i^- = (2a\mathbf{e}_i, 0).$$

Thus $\mathcal{S}$, and hence C , spans $\mathbb{R}^3 \oplus \mathbb{R}$. Since C is a convex cone, every real linear combination can be separated into its positive and negative parts, proving Eq. (149). $\square$

Proposition A.2 (Unique linear extension). *There exists a unique real-linear map*

$$\tilde{\mathcal{R}} : V \longrightarrow W \quad (150)$$

whose restriction to C is $\mathcal{R}$.

Proof. By Lemma A.1, write $Z = Y_+ - Y_-$ with $Y_\pm \in C$ and define

$$\tilde{\mathcal{R}}(Z) := \mathcal{R}(Y_+) - \mathcal{R}(Y_-). \quad (151)$$

If also $Z = Y'_+ - Y'_-$, then $Y_+ + Y'_- = Y'_+ + Y_-$. Additivity of $\mathcal{R}$ therefore gives

$$\mathcal{R}(Y_+) - \mathcal{R}(Y_-) = \mathcal{R}(Y'_+) - \mathcal{R}(Y'_-),$$

so the definition is independent of the decomposition. Additivity and positive homogeneity of $\mathcal{R}$ imply additivity and real homogeneity of $\tilde{\mathcal{R}}$. Any other linear extension must have the same value on every difference $Y_+ - Y_-$, proving uniqueness. $\square$

We next impose spatial covariance. The rotation group acts on both spaces by

$$R \oplus 1, \quad R \in SO(3),$$

and we require

$$\tilde{\mathcal{R}}(R\mathbf{D}, S) = (R \oplus 1)\tilde{\mathcal{R}}(\mathbf{D}, S). \quad (152)$$

Proposition A.3 (Rotational form of the readout). *Every real-linear map satisfying Eq. (152) has the form*

$$\tilde{\mathcal{R}}(\mathbf{D}, S) = (\alpha\mathbf{D}, \beta S) \quad (153)$$

for constants $\alpha, \beta \in \mathbb{R}$.

Proof. Write

$$\tilde{\mathcal{R}}(\mathbf{D}, S) = (A\mathbf{D} + \mathbf{b}S, \mathbf{c} \cdot \mathbf{D} + \beta S).$$

Equivariance gives $AR = RA$ for every $R \in SO(3)$, hence $A = \alpha I$. It also requires $R\mathbf{b} = \mathbf{b}$ and $R^T \mathbf{c} = \mathbf{c}$ for every rotation, so $\mathbf{b} = \mathbf{c} = \mathbf{0}$. $\square$

The operational assumptions of Sec. 2 select $\alpha, \beta > 0$. For a pure process ray

$$Y = \lambda(\mathbf{D}, S), \quad \lambda > 0,$$

the readout is therefore

$$(d\mathbf{x}, d\chi) = \lambda(\alpha\mathbf{D}, \beta S). \quad (154)$$

Defining

$$c := \frac{\alpha}{\beta}, \quad dt := \beta\lambda \quad (155)$$

gives

$$\left(\frac{d\mathbf{x}}{c}, d\chi\right) = dt(\mathbf{D}, S). \quad (156)$$

Equation (156) is a calibration of the two irreducible readout channels, not part of the abstract extension theorem. In particular, the observer duration t is introduced only after the additive cone map has been extended and its rotational form has been determined.

B Convex-Dual Examples and Newtonian Rest Geometry

This appendix gives explicit examples for the forward convex-dual construction used in Sec. 3. The underlying Legendre–Fenchel and support-function results are standard [10, 12].

For $r > 1$, define

$$K_r^+ := \{(\mathbf{D}, S) : |\mathbf{D}|^r + S^r \leq 1, S \geq 0\}. \quad (157)$$

Its vector projection is the Euclidean unit ball and its upper boundary is

$$\sigma_r(\mathbf{D}) = (1 - |\mathbf{D}|^r)^{1/r}, \quad |\mathbf{D}| \leq 1. \quad (158)$$

Let

$$r' := \frac{r}{r-1}, \quad \frac{1}{r} + \frac{1}{r'} = 1. \quad (159)$$

Proposition B.1. For $Q \geq 0$,

$$h_{K_r^+}(\mathbf{p}, Q) = (|\mathbf{p}|^{r'} + Q^{r'})^{1/r'}, \quad (160)$$

and therefore

$$H_{r,Q}(\mathbf{p}) = c (|\mathbf{p}|^{r'} + Q^{r'})^{1/r'}. \quad (161)$$

Proof. For $(\mathbf{D}, S) \in K_r^+$, Cauchy–Schwarz followed by Hölder’s inequality gives

$$\begin{aligned} \mathbf{p} \cdot \mathbf{D} + QS &\leq |\mathbf{p}| |\mathbf{D}| + QS \\ &\leq (|\mathbf{p}|^{r'} + Q^{r'})^{1/r'} (|\mathbf{D}|^r + S^r)^{1/r} \\ &\leq (|\mathbf{p}|^{r'} + Q^{r'})^{1/r'}. \end{aligned} \quad (162)$$

For $\mathbf{p} \neq 0$ and $Q > 0$, equality is attained by taking

$$\mathbf{D} = \frac{|\mathbf{p}|^{r'-1}}{(|\mathbf{p}|^{r'} + Q^{r'})^{1/r}} \frac{\mathbf{p}}{|\mathbf{p}|}, \quad S = \frac{Q^{r'-1}}{(|\mathbf{p}|^{r'} + Q^{r'})^{1/r}}.$$

The cases $\mathbf{p} = 0$ or $Q = 0$ follow directly or by continuity. Equation (161) then follows from $H_{K,Q} = c h_{K_r^+}(\mathbf{p}, Q)$. $\square$

For $Q > 0$, the corresponding extended-real Lagrangian is

$$L_{r,Q}(\mathbf{v}) = \begin{cases} -cQ \left[1 - \left(\frac{|\mathbf{v}|}{c} \right)^r \right]^{1/r}, & |\mathbf{v}| \leq c, \\ +\infty, & |\mathbf{v}| > c. \end{cases} \quad (163)$$

The boundary $|\mathbf{v}| = c$ belongs to the closed convex extension; for fixed $Q > 0$ it is not a finite-momentum massive state.

The Euclidean state body is the case $r = r' = 2$, for which

$$H_{2,Q}(\mathbf{p}) = c \sqrt{|\mathbf{p}|^2 + Q^2}.$$

For $p := |\mathbf{p}| \ll Q$,

$$H_{r,Q}(p) = cQ + \frac{cQ}{r'} \left(\frac{p}{Q} \right)^{r'} + O \left[cQ \left(\frac{p}{Q} \right)^{2r'} \right]. \quad (164)$$

With $m = Q/c$, the standard Newtonian behavior is

$$H(p) = mc^2 + \frac{p^2}{2m} + o(p^2). \quad (165)$$

Within the K_r^+ family this requires $r' = 2$, and hence

$$r = 2. \quad (166)$$

For $r > 2$, one has $1 < r' < 2$ and the leading correction is lower than quadratic. For $1 < r < 2$, one has $r' > 2$ and the quadratic kinetic coefficient vanishes. Thus the conventional nondegenerate Newtonian rest behavior selects the Euclidean member of this particular family.

This conclusion is only family-specific. For a general smooth compact state body, the Newtonian limit constrains only the local curvature of the upper boundary.

Proposition B.2 (Newtonian rest-curvature condition). *Suppose the upper boundary of a compact state body is*

$$S = \sigma(\mathbf{D})$$

near $\mathbf{D} = \mathbf{0}$, with $\sigma \in C^2$,

$$\sigma(\mathbf{0}) = 1, \quad \nabla \sigma(\mathbf{0}) = \mathbf{0}.$$

For

$$L_Q(\mathbf{v}) = -cQ\sigma\left(\frac{\mathbf{v}}{c}\right), \quad (167)$$

the isotropic Newtonian expansion

$$L_Q(\mathbf{v}) = -mc^2 + \frac{1}{2}m|\mathbf{v}|^2 + o(|\mathbf{v}|^2), \quad m = \frac{Q}{c}, \quad (168)$$

holds if and only if

$$D^2\sigma(\mathbf{0}) = -I. \quad (169)$$

Proof. Taylor expansion gives

$$\sigma(\mathbf{D}) = 1 + \frac{1}{2} \mathbf{D}^T D^2 \sigma(\mathbf{0}) \mathbf{D} + o(|\mathbf{D}|^2).$$

Substitution of $\mathbf{D} = \mathbf{v}/c$ gives

$$L_Q(\mathbf{v}) = -cQ - \frac{Q}{2c} \mathbf{v}^T D^2 \sigma(\mathbf{0}) \mathbf{v} + o(|\mathbf{v}|^2). \quad (170)$$

Since $m = Q/c$, comparison with Eq. (168) is equivalent to $D^2 \sigma(\mathbf{0}) = -I$. $\square$

For a rotationally invariant state body, $D^2 \sigma(\mathbf{0}) = -\lambda I$; the Newtonian normalization fixes $\lambda = 1$. This is only a local second-order condition. It does not imply globally that

$$\sigma(\mathbf{D}) = \sqrt{1 - |\mathbf{D}|^2}.$$

Infinitely many compact convex bodies can agree with the Euclidean half-ball to second order at the static point and differ elsewhere.

The scale-free sector contains still less information. Setting $Q = 0$ in Eq. (161) gives

$$H_{r,0}(\mathbf{p}) = c|\mathbf{p}| \quad (171)$$

for every $r > 1$. More generally, if Ω is the vector projection of a compact state body K , then

$$H_{K,0}(\mathbf{p}) = c h_K(\mathbf{p}, 0) = c h_\Omega(\mathbf{p}). \quad (172)$$

Hence the $Q = 0$ dispersion probes only the projected velocity geometry, whereas the massive sector $Q > 0$ also probes the scalar profile of the state body.

For the Euclidean body,

$$\sigma_2(\mathbf{D}) = \sqrt{1 - |\mathbf{D}|^2}, \quad D^2 \sigma_2(\mathbf{0}) = -I,$$

and

$$H_{2,Q}(\mathbf{p}) = c \sqrt{|\mathbf{p}|^2 + Q^2}.$$

Thus the Newtonian rest limit distinguishes the Euclidean member within the explicit K_r^+ family, but does not by itself establish global Euclidean state geometry in the general case.

C Degenerate Solutions and MS-Invisible Canonical Directions

The pointwise MS consistency equation admits smooth convex solutions that fail both representation completeness and Legendre nondegeneracy. This appendix analyzes the projection family introduced in Sec. 3.

Let

$$\mathbf{R}^2 = \mathbf{R}, \quad \mathbf{R}^T = \mathbf{R} \quad (173)$$

be an orthogonal projection on $\mathbb{R}^3$, with

$$k := \text{rank } \mathbf{R}.$$

Decompose

$$\mathbf{p} = \mathbf{p}_\parallel + \mathbf{p}_\perp, \quad \mathbf{p}_\parallel := \mathbf{R}\mathbf{p}, \quad \mathbf{p}_\perp := (\mathbf{I} - \mathbf{R})\mathbf{p}. \quad (174)$$

Define

$$H_{\mathbf{R}}(\mathbf{p}) := c \sqrt{Q^2 + |\mathbf{p}_\parallel|^2}, \quad Q > 0. \quad (175)$$

Proposition C.1 (Projection solutions). *The Hamiltonian H_R is smooth and convex and satisfies*

$$H_R(\mathbf{0}) = cQ \quad (176)$$

and

$$\frac{|\nabla H_R|^2}{c^2} + \frac{(H_R - \mathbf{p} \cdot \nabla H_R)^2}{c^2 Q^2} = 1, \quad H_R - \mathbf{p} \cdot \nabla H_R > 0. \quad (177)$$

Proof. The function $\mathbf{z} \mapsto c \sqrt{Q^2 + |\mathbf{z}|^2}$ is smooth and convex for $Q > 0$, and H_R is its pullback by the linear map R . Moreover,

$$\nabla H_R = \frac{c \mathbf{p}_{\parallel}}{\sqrt{Q^2 + |\mathbf{p}_{\parallel}|^2}}, \quad (178)$$

and, since $\mathbf{p} \cdot \mathbf{p}_{\parallel} = |\mathbf{p}_{\parallel}|^2$,

$$H_R - \mathbf{p} \cdot \nabla H_R = \frac{cQ^2}{\sqrt{Q^2 + |\mathbf{p}_{\parallel}|^2}}. \quad (179)$$

Substitution gives Eq. (177). $\square$

The corresponding MS state is

$$\mathbf{D}_R = \frac{\mathbf{p}_{\parallel}}{\sqrt{Q^2 + |\mathbf{p}_{\parallel}|^2}}, \quad S_R = \frac{Q}{\sqrt{Q^2 + |\mathbf{p}_{\parallel}|^2}}. \quad (180)$$

It depends only on $\mathbf{p}_{\parallel}$.

Proposition C.2 (Canonical image). *The canonical state map generated by H_R has image*

$$\Psi_R(\mathbb{R}^3) = \mathcal{S}_R := \{(\mathbf{D}, S) \in \mathcal{S} : \mathbf{D} \in \text{Ran } R\}, \quad (181)$$

or equivalently

$$\nabla H_R(\mathbb{R}^3) = \{\mathbf{v} \in \text{Ran } R : |\mathbf{v}| < c\}. \quad (182)$$

Hence the image has dimension $k = \text{rank } R$.

Proof. Equation (178) gives the inclusion

$$\nabla H_R(\mathbb{R}^3) \subset \{\mathbf{v} \in \text{Ran } R : |\mathbf{v}| < c\}.$$

Conversely, for any $\mathbf{v} \in \text{Ran } R$ with $|\mathbf{v}| < c$, choose

$$\mathbf{p}_{\parallel} = \frac{Q\mathbf{v}}{\sqrt{c^2 - |\mathbf{v}|^2}}. \quad (183)$$

Then $\nabla H_R = \mathbf{v}$. The scalar component follows from $S = \sqrt{1 - |\mathbf{D}|^2}$ on the positive branch. $\square$

Thus $k = 3$ is the representation-complete case within this family; for $k < 3$, the canonical image is a lower-dimensional section of the massive MS manifold.

The differential degeneracy is equally explicit:

$$\nabla^2 H_R = c \left[\frac{R}{\sqrt{Q^2 + |\mathbf{p}_{\parallel}|^2}} - \frac{\mathbf{p}_{\parallel} \otimes \mathbf{p}_{\parallel}}{(Q^2 + |\mathbf{p}_{\parallel}|^2)^{3/2}} \right]. \quad (184)$$

Proposition C.3 (Exact Hessian kernel). *For every $\mathbf{p}$,*

$$\ker \nabla^2 H_R(\mathbf{p}) = \ker \mathbf{R}. \quad (185)$$

In particular, $\text{rank } \nabla^2 H_R = k$ everywhere.

Proof. Every $\mathbf{w} \in \ker \mathbf{R}$ satisfies $\mathbf{R}\mathbf{w} = 0$ and $\mathbf{p}_\parallel \cdot \mathbf{w} = 0$, so Eq. (184) gives $\nabla^2 H_R \mathbf{w} = 0$.

On the other hand, for $\mathbf{z} \in \text{Ran } \mathbf{R}$,

$$\begin{aligned} \mathbf{z}^T \nabla^2 H_R \mathbf{z} &= \frac{c}{\sqrt{Q^2 + |\mathbf{p}_\parallel|^2}} \left[|\mathbf{z}|^2 - \frac{(\mathbf{p}_\parallel \cdot \mathbf{z})^2}{Q^2 + |\mathbf{p}_\parallel|^2} \right] \\ &\geq \frac{cQ^2}{(Q^2 + |\mathbf{p}_\parallel|^2)^{3/2}} |\mathbf{z}|^2. \end{aligned} \quad (186)$$

This is positive for nonzero $\mathbf{z}$, proving the claim. $\square$

The degeneracy persists along the affine leaves

$$\mathbf{p} + \ker \mathbf{R}. \quad (187)$$

Indeed, for every $\mathbf{w} \in \ker \mathbf{R}$,

$$H_R(\mathbf{p} + \mathbf{w}) = H_R(\mathbf{p}), \quad \Psi_R(\mathbf{p} + \mathbf{w}) = \Psi_R(\mathbf{p}). \quad (188)$$

Thus the dependence of both the Hamiltonian and the MS state map factors through

$$\mathbb{R}^3 / \ker \mathbf{R} \simeq \text{Ran } \mathbf{R}. \quad (189)$$

On this quotient their functional dependence is the standard lower-dimensional massive square root,

$$H_R = c \sqrt{Q^2 + |\mathbf{p}_\parallel|^2}.$$

No claim is made that $\ker \mathbf{R}$ is a gauge redundancy of the full canonical phase space; the statement is only that these directions are invisible to the assumed MS state representation.

This distinction is also visible in the generator vector. For a general Hamiltonian we introduced

$$P_H = \left(\frac{H}{c}, \mathbf{p} \right). \quad (190)$$

For the projection solution,

$$\eta(P_H, P_H) = -Q^2 + |\mathbf{p}_\perp|^2. \quad (191)$$

Thus P_H does not in general lie on the standard massive shell when $\mathbf{R} \neq I$.

The projected vector

$$P_R^{\text{vis}} := \left(\frac{H_R}{c}, \mathbf{p}_\parallel \right) \quad (192)$$

does satisfy

$$\eta(P_R^{\text{vis}}, P_R^{\text{vis}}) = -Q^2. \quad (193)$$

This vector records only the generator component visible to the MS state; it is not asserted to replace the full canonical generator of three-dimensional translations.

Consequently, the pointwise master equation constrains the combination

$$\nabla H, \quad H - \mathbf{p} \cdot \nabla H,$$

but does not by itself guarantee that every canonical momentum direction is faithfully represented in the MS state. In the projection family,

$$\mathbf{R} = I \iff \Psi_{\mathbf{R}}(\mathbb{R}^3) = \mathcal{S} \iff \nabla^2 H_{\mathbf{R}} \text{ is nondegenerate everywhere.} \quad (194)$$

Equivalently, $\mathbf{R} = I$ is the unique member for which the full vector P_H lies on $\eta(P_H, P_H) = -Q^2$ for every canonical momentum.

The projection family therefore gives explicit smooth convex counterexamples to rigidity based on the master equation alone. Representation completeness excludes missing MS motion directions, while dense Hessian nondegeneracy excludes persistent canonical directions that the MS representation does not detect. No classification of all globally degenerate solutions is attempted here.

D Internal Reference Realization of Observer Duration

The operational construction of Sec. 2 assigns to a massive process segment a positive radial duration dt in a chosen inertial operational representation. In the main text this duration is observer dependent, whereas the scalar accumulation $d\chi$ is attached to the physical process segment itself. The purpose of this appendix is to record a fully internal realization of the observer label: the operational reference may itself be taken to be a physical process belonging to the same closed system. This interpretation does not modify the compact-state construction, the Hamiltonian rigidity problem, or the observer-covariance results of the main text.

Let $\mathcal{U}$ be a closed physical system containing the massive process A under consideration and an admissible inertial reference process O . The introduction of O is not an enlargement of the free-particle Hamiltonian problem to a two-particle dynamical system; O serves only to realize internally the operational reference already present in Sec. 2.

For a finite positively oriented segment σ_I of a massive process I , let

$$\Delta\chi_I(\sigma_I) > 0$$

denote its intrinsic scalar accumulation. The finite-segment notation is used only to make process composition explicit. Additivity means

$$\Delta\chi_I(\sigma_2 \circ \sigma_1) = \Delta\chi_I(\sigma_1) + \Delta\chi_I(\sigma_2) \quad (195)$$

whenever the two segments are composable.

To represent the duration assigned by O without introducing an external clock, suppose that the operational comparison between A and O determines a segment-matching map

$$\mathfrak{m}_O^A : \text{Seg}(A) \longrightarrow \text{Seg}(O). \quad (196)$$

We require that $\mathfrak{m}_O^A$ map every nontrivial positively oriented segment of A to a nontrivial positively oriented segment of O , and that it preserve concatenation:

$$\mathfrak{m}_O^A(\sigma_2 \circ \sigma_1) = \mathfrak{m}_O^A(\sigma_2) \circ \mathfrak{m}_O^A(\sigma_1). \quad (197)$$

For self-comparison we take

$$m_O^O = \text{id}_{\text{Seg}(O)}. \quad (198)$$

We also assume sufficient local regularity for the resulting finite-segment assignment to admit the infinitesimal limit used below.

The duration assigned by the internal reference process O to a segment σ_A is defined by

$$\Delta t_O(\sigma_A) := \Delta \chi_O(m_O^A(\sigma_A)). \quad (199)$$

This definition immediately gives both positivity and additivity. Indeed,

$$\begin{aligned} \Delta t_O(\sigma_2 \circ \sigma_1) &= \Delta \chi_O(m_O^A(\sigma_2 \circ \sigma_1)) \\ &= \Delta \chi_O(m_O^A(\sigma_2) \circ m_O^A(\sigma_1)) \\ &= \Delta t_O(\sigma_1) + \Delta t_O(\sigma_2). \end{aligned} \quad (200)$$

For self-comparison, Eq. (198) gives

$$\Delta t_O(\sigma_O) = \Delta \chi_O(\sigma_O). \quad (201)$$

In the local limit, denote the corresponding reference duration by $dt_O(A) > 0$, and let

$$d\mathbf{r}_{A|O}$$

be the infinitesimal spatial displacement of A in the three-dimensional Euclidean spatial readout associated with the inertial representation defined by O . No absolute spatial position or absolute velocity is introduced. If the radial duration appearing in the operational MS realization of Sec. 2 is implemented by this internal reference construction, then restoring the observer label gives

$$\left(\frac{d\mathbf{r}_{A|O}}{c}, d\chi_A \right) = dt_O(A) (\mathbf{D}_{A|O}, S_{A|O}), \quad (202)$$

with

$$|\mathbf{D}_{A|O}|^2 + S_{A|O}^2 = 1, \quad S_{A|O} > 0. \quad (203)$$

Hence

$$\mathbf{D}_{A|O} = \frac{d\mathbf{r}_{A|O}}{c dt_O(A)}, \quad S_{A|O} = \frac{d\chi_A}{dt_O(A)}. \quad (204)$$

The compact-state normalization therefore yields

$$c^2 d\chi_A^2 = c^2 dt_O(A)^2 - |d\mathbf{r}_{A|O}|^2. \quad (205)$$

Introducing the velocity assigned by O ,

$$\mathbf{v}_{A|O} := \frac{d\mathbf{r}_{A|O}}{dt_O(A)}, \quad (206)$$

one obtains

$$\mathbf{D}_{A|O} = \frac{\mathbf{v}_{A|O}}{c}, \quad S_{A|O} = \frac{d\chi_A}{dt_O(A)} = \sqrt{1 - \frac{|\mathbf{v}_{A|O}|^2}{c^2}}. \quad (207)$$

For the reference process in its own comoving inertial representation,

$$d\mathbf{r}_{O|O} = 0.$$

Equation (201) then gives locally

$$dt_O(O) = d\chi_O,$$

and the MS normalization implies

$$\mathbf{D}_{O|O} = \mathbf{0}, \quad S_{O|O} = 1. \quad (208)$$

Thus the pure-staticity calibration of Eq. (36) is recovered entirely within the closed system: the reference process realizes its own radial duration through its intrinsic scalar accumulation.

Equation (205) is the observer-labeled form of Eq. (39). As in the main text, the proper-time interpretation is introduced only after this Lorentzian quadratic relation has been obtained:

$$d\tau_A := d\chi_A. \quad (209)$$

Accordingly, $d\chi_A$ is intrinsic to the massive process segment, whereas $dt_O(A)$ is the duration assigned to that segment relative to the chosen internal reference process.

The construction is compatible with the observer covariance derived in Sec. 4. If two admissible inertial reference processes O and O' describe the same physical segment of A , define

$$dX_{A|O} := (c dt_O(A), d\mathbf{r}_{A|O}), \quad dX_{A|O'} := (c dt_{O'}(A), d\mathbf{r}_{A|O'}). \quad (210)$$

Under the hypotheses of Sec. 4,

$$dX_{A|O'} = \Lambda_{O'O} dX_{A|O}, \quad \Lambda_{O'O} \in O^\uparrow(1, 3), \quad (211)$$

and for reference changes continuously connected to the identity,

$$\Lambda_{O'O} \in SO^+(1, 3). \quad (212)$$

Since $d\chi_A$ is attached to the physical process segment rather than to a particular inertial representation, the two descriptions satisfy

$$\eta(dX_{A|O}, dX_{A|O}) = \eta(dX_{A|O'}, dX_{A|O'}) = -c^2 d\chi_A^2, \quad \eta = \text{diag}(-1, +1, +1, +1). \quad (213)$$

Thus the internal-reference construction supplies an interpretation of the observer label while leaving the Lorentz-covariance argument of the main text unchanged.

As a simple consistency check, let the process A itself be an inertial reference process O' moving with constant relational velocity $\mathbf{u}$ in the representation associated with O . Then

$$d\mathbf{r}_{O'|O} = \mathbf{u} dt_O(O'),$$

and Eq. (205) gives

$$d\chi_{O'} = dt_O(O') \sqrt{1 - \frac{|\mathbf{u}|^2}{c^2}}. \quad (214)$$

In the comoving representation of O' ,

$$dt_{O'}(O') = d\chi_{O'},$$

so that

$$dt_{O'}(O') = dt_O(O') \sqrt{1 - \frac{|\mathbf{u}|^2}{c^2}}. \quad (215)$$

This is the usual inertial time-dilation relation expressed entirely in terms of process accumulation, relational spatial displacement, and internal reference duration.

It is important that the two reference matchings used when the roles of O and O' are interchanged need not associate the same pairs of spatially separated endpoints. They encode the corresponding operational simultaneity assignments. Thus quantities such as

$$\frac{d\chi_{O'}}{dt_O(O')} \quad \text{and} \quad \frac{d\chi_O}{dt_{O'}(O')}$$

are not to be interpreted as reciprocal ratios constructed from one fixed pair of process segments. Their mutual consistency is instead governed by the Lorentz transformation Eq. (211).

Finally, the internal-reference interpretation has a limiting case that clarifies its scope. If the closed system contains only one completely structureless distinguishable process channel O , then only self-comparison is available. In that case

$$d\mathbf{r}_{O|O} = 0, \quad (\mathbf{D}_{O|O}, S_{O|O}) = (\mathbf{0}, 1), \quad (216)$$

and no nonzero relational velocity is operationally defined. A nontrivial realization of the motion sector therefore requires sufficient distinguishable relational structure within the closed system. Such structure need not correspond to an additional particle species or to an observer external to the system; it is enough that there exist physically distinguishable process channels capable of serving as mutual operational references.

The internal construction above should therefore be understood as a realization of the observer structure already used in the main text, not as an additional derivation of temporal ordering or of the existence of the intrinsic accumulation $d\chi$. In particular, the segment matching $\mathfrak{m}_O^A$ is an operational realizability condition and is not implied by the compact-state normalization $|\mathbf{D}|^2 + S^2 = 1$. The temporal assumptions of Sec. 5 are consequently unchanged. What the present appendix establishes is that the observer-dependent radial duration of the MS construction can be implemented without introducing a clock or reference system external to the closed physical system.

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